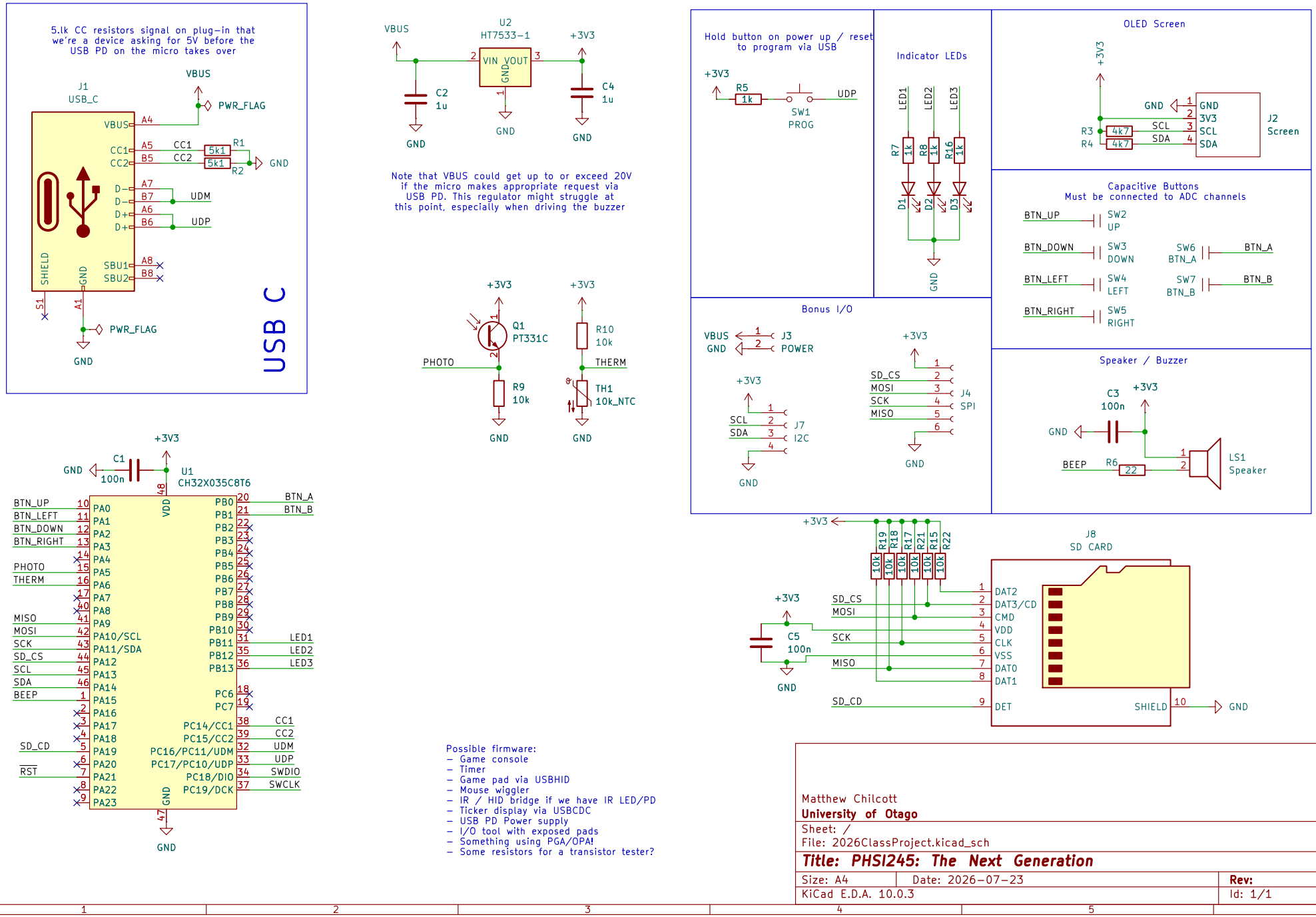

5.1k CC resistors signal on plug-in that we're a device asking for 5V before the USB PD on the micro takes over
J1
USB_C
VBUS
PWR_FLAG
CC1
CC2
5k1
R1
R2
GND
UDM
UDP
SHIELD
SBU1
SBU2
S1
A1
USB C
U2
HT7533-1
VIN
VOUT
C2
1u
C4
1u
Note that VBUS could get up to or exceed 20V if the micro makes appropriate request via USB PD. This regulator might struggle at this point, especially when driving the buzzer
Hold button on power up / reset to program via USB
R5
1k
SW1
PROG
Indicator LEDs
LED1
LED2
LED3
R7
R8
R16
D1
D2
D3
OLED Screen
J2
Screen
SCL
SDA
3V3
R3
R4
4k7
Capacitive Buttons
Must be connected to ADC channels
BTN_UP
BTN_DOWN
BTN_LEFT
BTN_RIGHT
SW2
UP
SW3
DOWN
SW4
LEFT
SW5
RIGHT
SW6
BTN_A
SW7
BTN_B
Bonus I/O
J3
POWER
J4
SPI
J7
I2C
Speaker / Buzzer
C3
100n
LS1
Speaker
BEEP
R6
22
Q1
PT331C
R9
10k
R10
10k
TH1
10k_NTC
PHOTO
THERM
U1
CH32X035C8T6
C1
100n
VDD
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10/SCL
PA11/SDA
PA12
PA13
PA14
PA15
PA16
PA17
PA18
PA19
PA20
PA21
PA22
PA23
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PC6
PC7
PC14/CC1
PC15/CC2
PC16/PC11/UDM
PC17/PC10/UDP
PC18/DIO
PC19/DCK
BTN_UP
BTN_LEFT
BTN_DOWN
BTN_RIGHT
PHOTO
THERM
MISO
MOSI
SCK
SD_CS
SCL
SDA
BEEP
SD_CD
RST
BTN_A
BTN_B
LED1
LED2
LED3
CC1
CC2
UDM
UDP
SWDIO
SWCLK
J8
SD CARD
R19
R18
R17
R21
R15
R22
10k
C5
100n
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET
SHIELD
Possible firmware:
– Game console
– Timer
– Game pad via USBHID
– Mouse wiggler
– IR / HID bridge if we have IR LED/PD
– Ticker display via USBCDC
– USB PD Power supply
– I/O tool with exposed pads
– Something using PGA/OPA!
– Some resistors for a transistor tester?
Matthew Chilcott
University of Otago
Sheet: /
File: 2026ClassProject.kicad_sch
Title: PHSI245: The Next Generation
Size: A4
Date: 2026-07-23
Rev:
KiCad E.D.A. 10.0.3
Id: 1/1